moving solid bodies would always have to retain their same shape and size relative to absolute space, regardless of how fast they moved.

Hendrik Lorentz in Amsterdam also believed the Michelson–Morley experiment, and he took seriously Fitzgerald's suggestion that moving objects contract. Fitzgerald, upon learning of this, wrote to Lorentz expressing delight, since "I have been rather laughed at for my view over here." In a search for deeper understanding, Lorentz—and independently Henri Poincaré in Paris, France, and Joseph Larmor in Cambridge, England—reexamined the laws of electromagnetism, and noticed a peculiarity that dovetailed with Fitzgerald's length-contraction idea:

If one expressed Maxwell's electromagnetic laws in terms of the electric and magnetic fields measured at rest in absolute space, the laws took on an especially simple and beautiful mathematical form. For example, one of the laws said, simply, "As seen by anyone at rest in absolute space, magnetic field lines have no ends" (see Figure 1.1a,b). However, if one expressed Maxwell's laws in terms of the slightly different fields measured by a moving person, then the laws looked far more complicated and ugly. In particular, the "no ends" law became, "As seen by someone in motion, most magnetic field lines are endless, but a few get cut by the motion, thereby acquiring ends. Moreover, when the moving person shakes the magnet, new field lines get cut, then heal, then get cut again, then reheal" (see Figure 1.lc).

The new mathematical discovery by Lorentz, Poincaré, and Larmor was a way to make the moving person's electromagnetic laws look beautiful, and in fact look identical to the laws used by a person at rest in absolute space: "Magnetic field lines never end, under any circumstances whatsoever." One could make the laws take on this beautiful form by pretending, contrary to Newtonian precepts, that all moving objects get contracted along their direction of motion by precisely the amount that Fitzgerald needed to explain the Michelson—Morley experiment!

If the Fitzgerald contraction had been the only "new physics" that one needed to make the electromagnetic laws universally simple and beautiful, Lorentz, Poincaré, and Larmor, with their intuitive faith that the laws of physics *ought to be* beautiful, might have cast aside Newtonian precepts and believed firmly in the contraction. However, the contraction by itself was not enough. To make the laws beautiful, one also had to pretend that